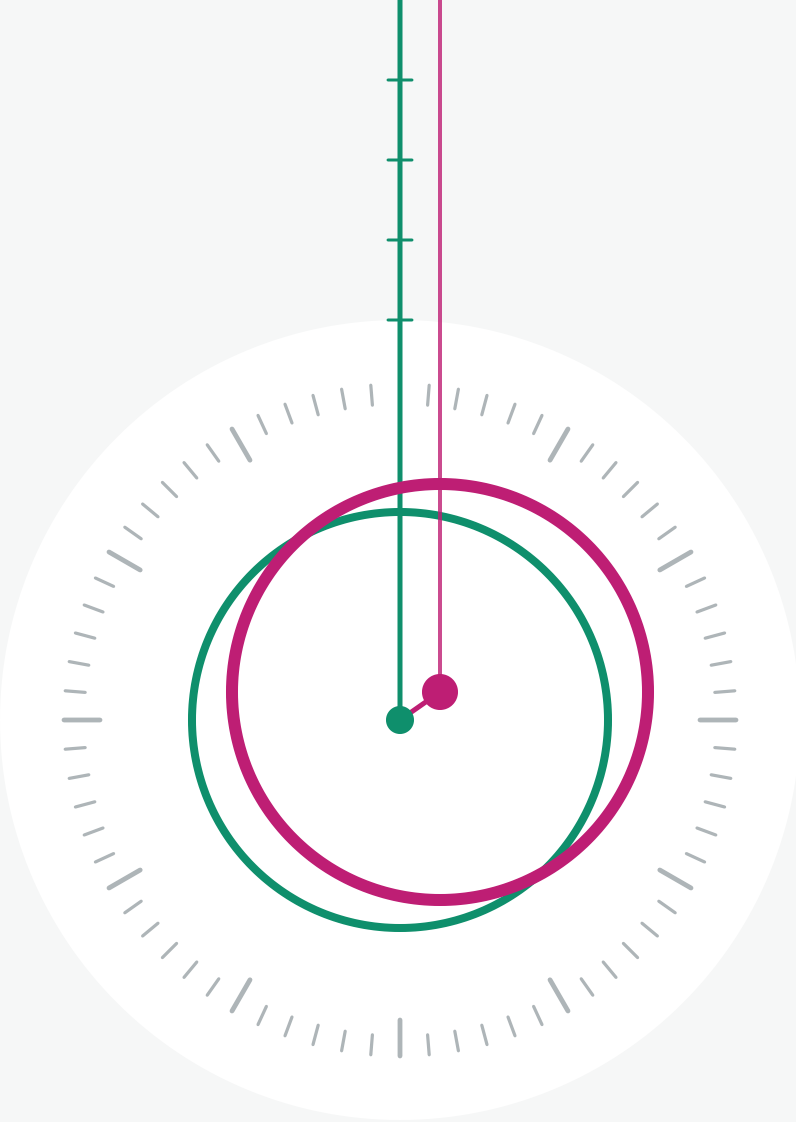
DATUM LINE

Confidence with controls.

Every lab’s cells, on one map.



Kat Titterton · San Francisco



macrofluidic.me/resume

Katherine Titterton

2023–2026 **Axiom**
founding MTS

2022–2023 **insitro**
neuroscience assays

2021–2022 **MIT**
expansion microscopy

2018–2021 **abbvie**
tau · neurons

2017–2018 **HARVARD MEDICAL SCHOOL**
Xenopus oocytes

2017 **COLGATE** BA
neuroscience + math biology

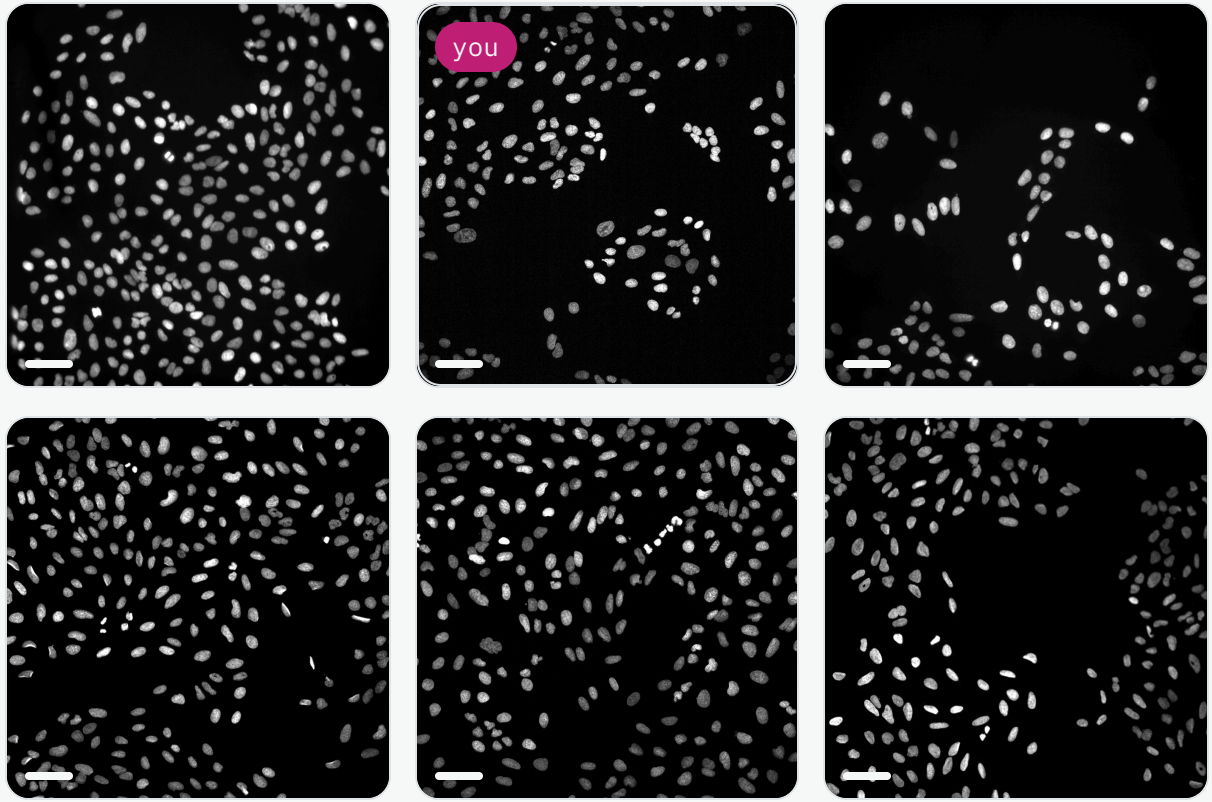
MY WORK

1. Cell Painting in primary human hepatocytes
   Cell Systems 2026 · Ewald, Broad Institute
2. Counting cells predicts bioactivity benchmarks
   Nat Commun 2026 · Seal, Univ. of Cambridge
3. Cell Painting of hepatotoxic botanicals
   bioRxiv 2026 · Ollitrault, EMBL-EBI
4. An antibody to pathological tau
   Mol Neurodegen Adv 2026 · Yanamandra, AbbVie
5. Capturing synaptic proteopathic seeds
   iScience 2026 · Sebastian, AbbVie
6. A neuronal tau aggregation assay
   J Biol Chem 2025 · Wu, AbbVie
7. Multiplexed imaging of brain nanostructures
   Nat Commun 2024 · Kang, MIT
8. Tau aggregation in human iPSC neurons
   iScience 2022 · Manos, AbbVie
9. Neurosteroid photolabeling of $GABA_A$ receptors
   Eur J Med Chem 2019 · Wu, U Illinois Chicago

Do my cells look “normal”?

asked @ five benches
… answered @ none

One lab, or six?



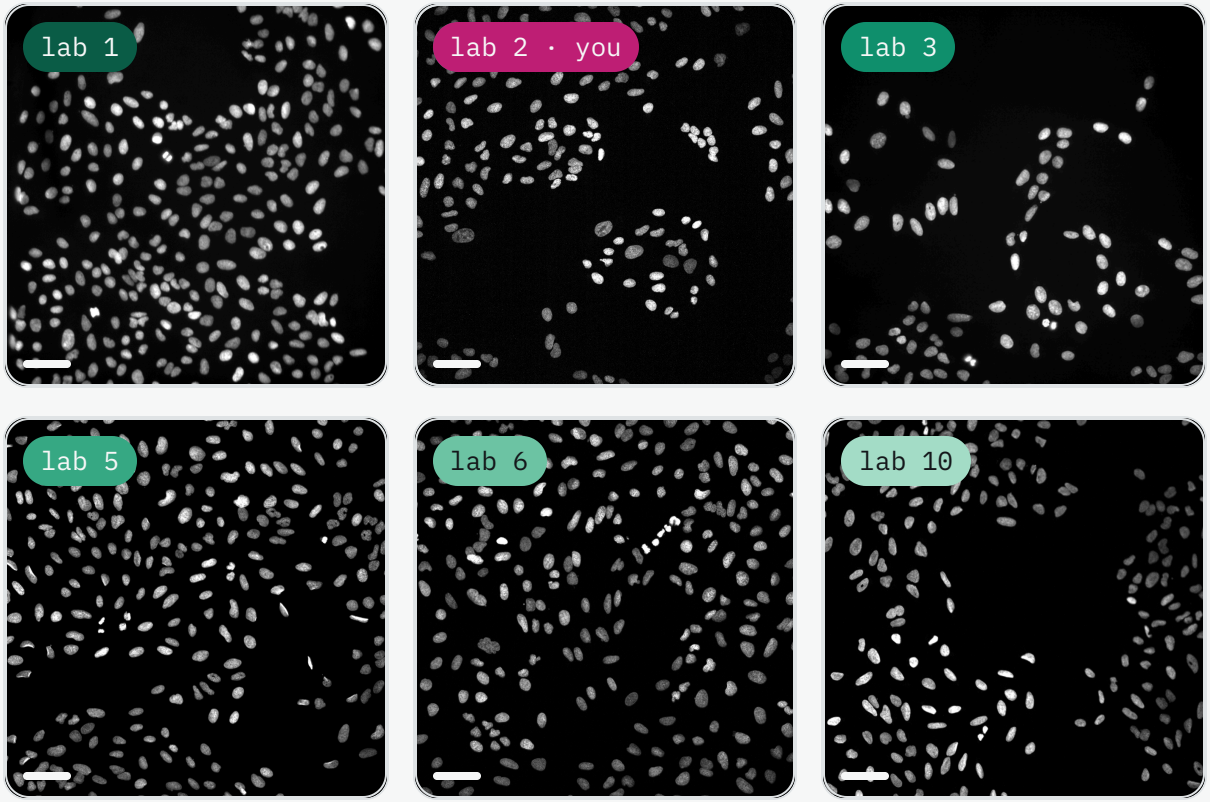
same cells ✓

same protocol ✓

no drug ✓

fig 1 Cell Painting · DNA channel · 93,228 wells · 11 labs · plates held out · bars 100 µm · JUMP [7] · A2, A3

Could you tell?
Classifiers can.

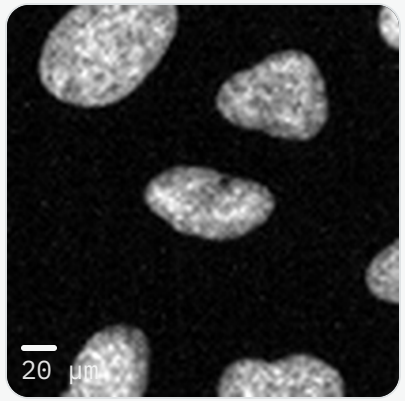


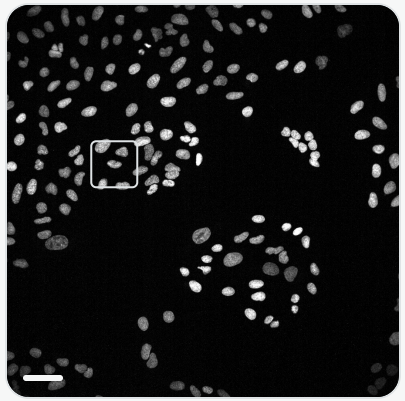
99.9% CLASSIFIER

9% CHANCE

fig 1 Cell Painting · DNA channel · 93,228 wells · 11 labs · plates held out · bars 100 µm · JUMP [7] · A2, A3

One well, onto the map.





3,773
numbers

one point

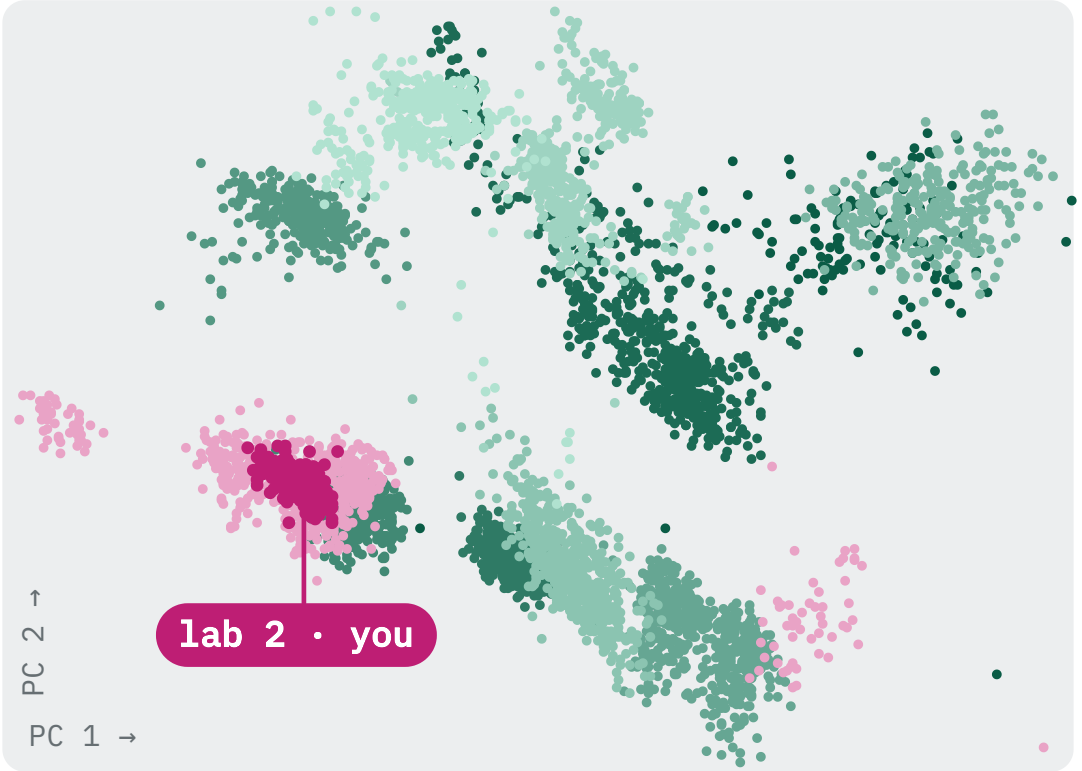


fig 1 Cell Painting · DNA channel · 93,228 wells · 11 labs · plates held out · bars 100 µm · JUMP [7] · A2, A3

Eleven labs.
Eleven “normals.”



BETWEEN LABS

6.2

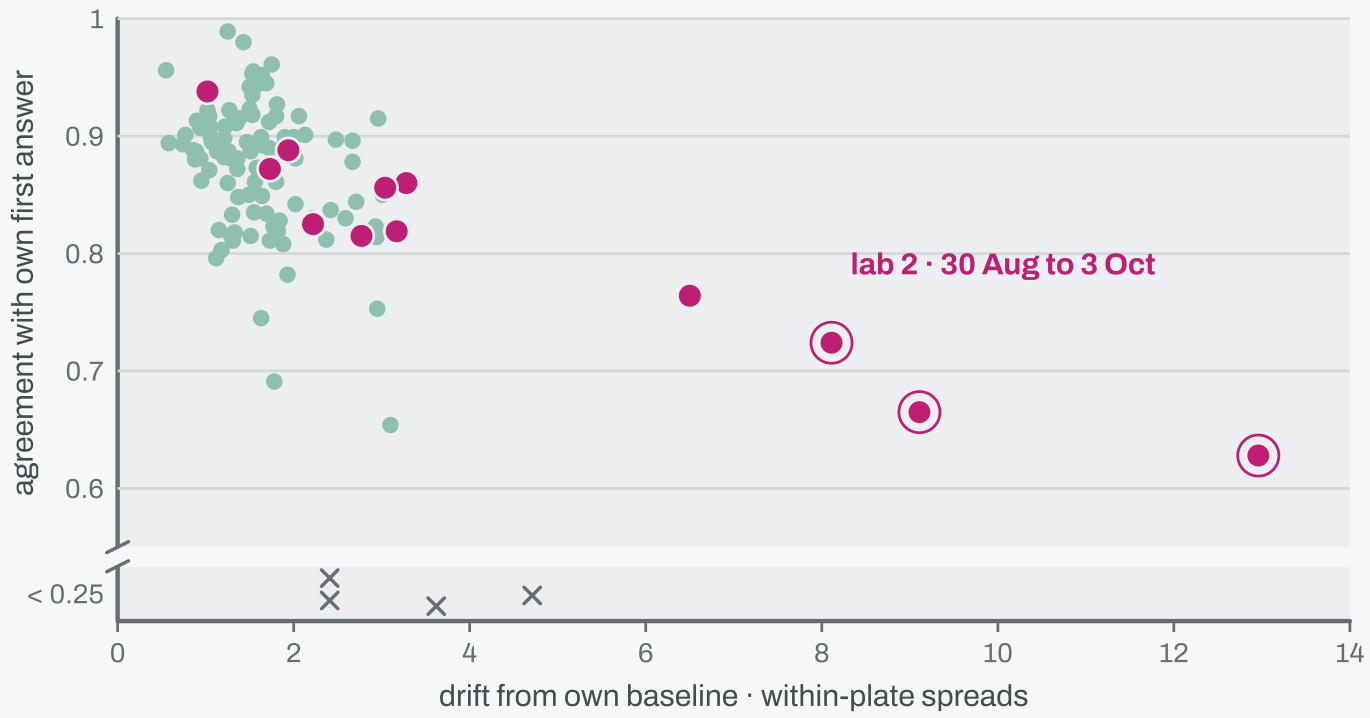
lab to lab

inside one plate

labs sit 6.2× farther apart
than wells on one plate

fig 2 15,352 untreated wells · PCA of 3,773 numbers · JUMP [7]

Where you sit predicts what you measure.



INSIDE LABS

ρ = −0.71

more drift · less agreement

LAB 2 · YOU · 30 AUG TO 3 OCT

drifted 3–5× its summer level

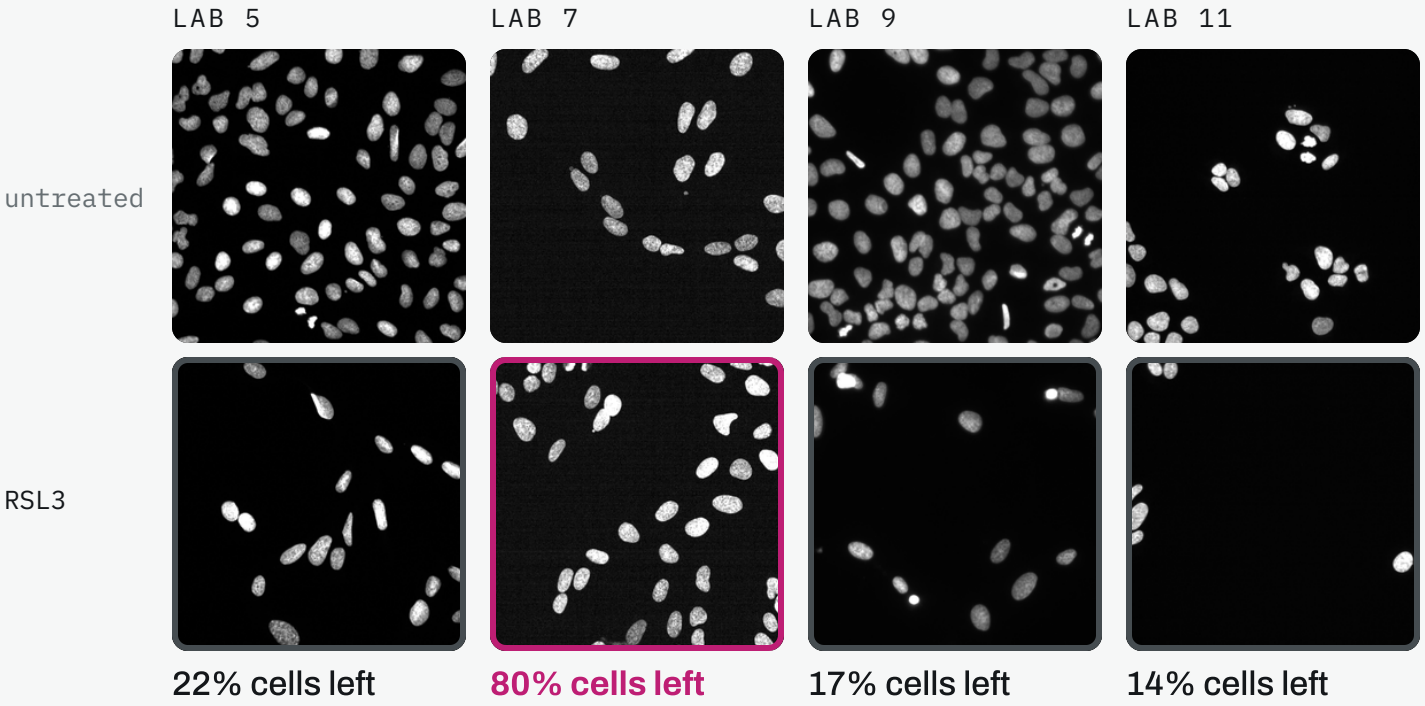
0.85 → 0.67

same drugs, matched against its own June answers

fig 3 118 batches · 11 labs · JUMP [7] · A8, A13

Does my control look like yours?

RSL3 · A STANDARD FERROPTOSIS CONTROL · IN ABOUT 1,200 PAPERS [16]



SAME DRUG · FOUR LABS

0.77 → 0.32

agreement, well to well · same lab → other labs
1 = same effect

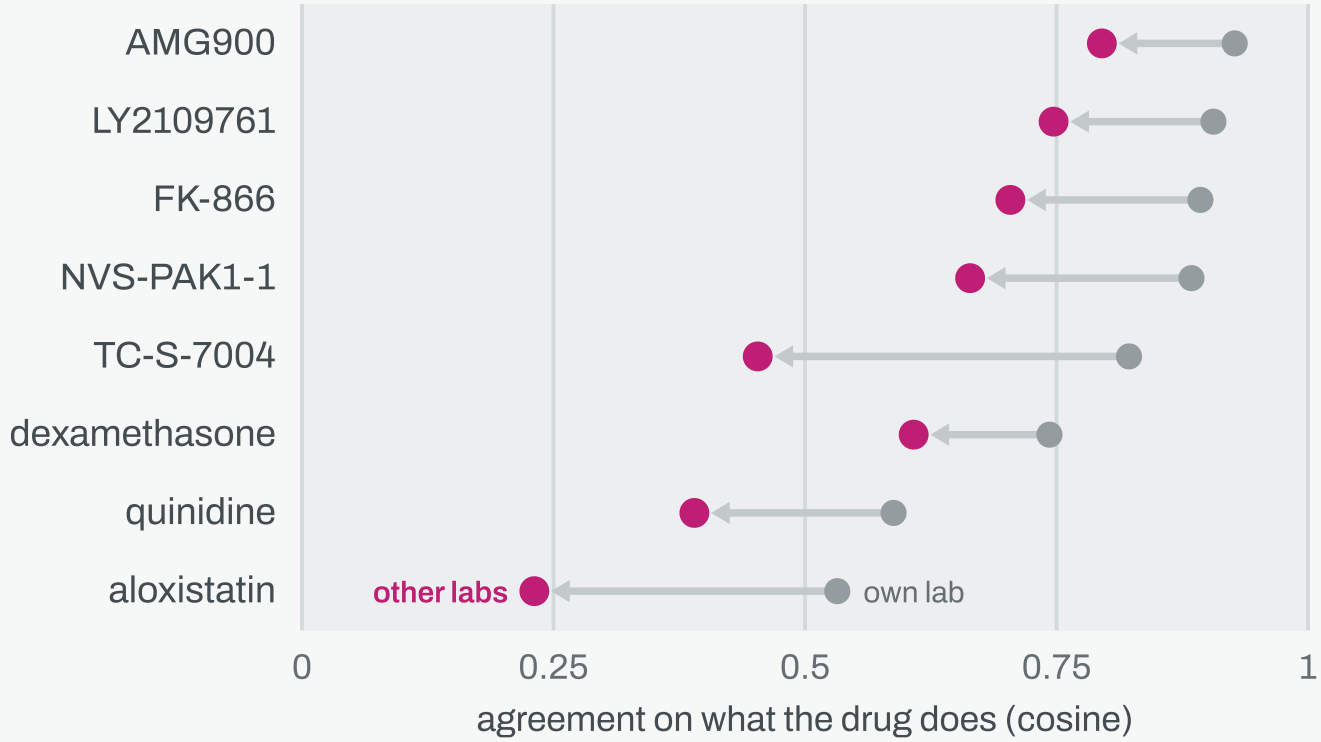
lab 7: **80%** of cells left
the others: 14–22%

lab 7's other controls worked:
the drug response differed, not the camera

one well per plate · 12 plates · small numbers, A9

fig 4a RSL3 · one well per plate, beside its untreated wells · DNA stain, 300 µm · JUMP [7] · PubMed [16]

Does my control look like yours?



0.79 → 0.57

all eight JUMP controls · own lab → other labs

ON THE MAP · PER BATCH

0.75 → 0.78

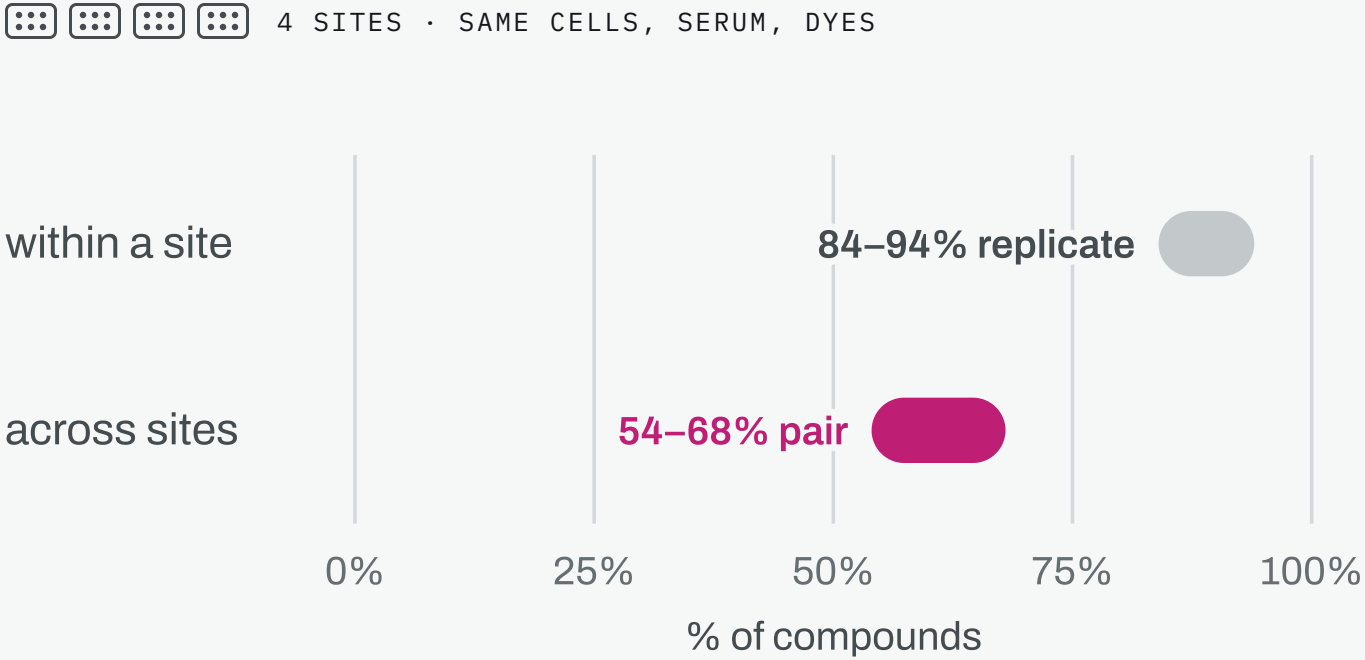
agreement with other labs · up in 121 of 125

0.52 → 0.59

drugs told apart · up in every batch

fig 4b 1,829 plates · 11 labs · JUMP [7] · A9, A14

Data is the new currency.
Lab A + lab B ≠ one dataset.



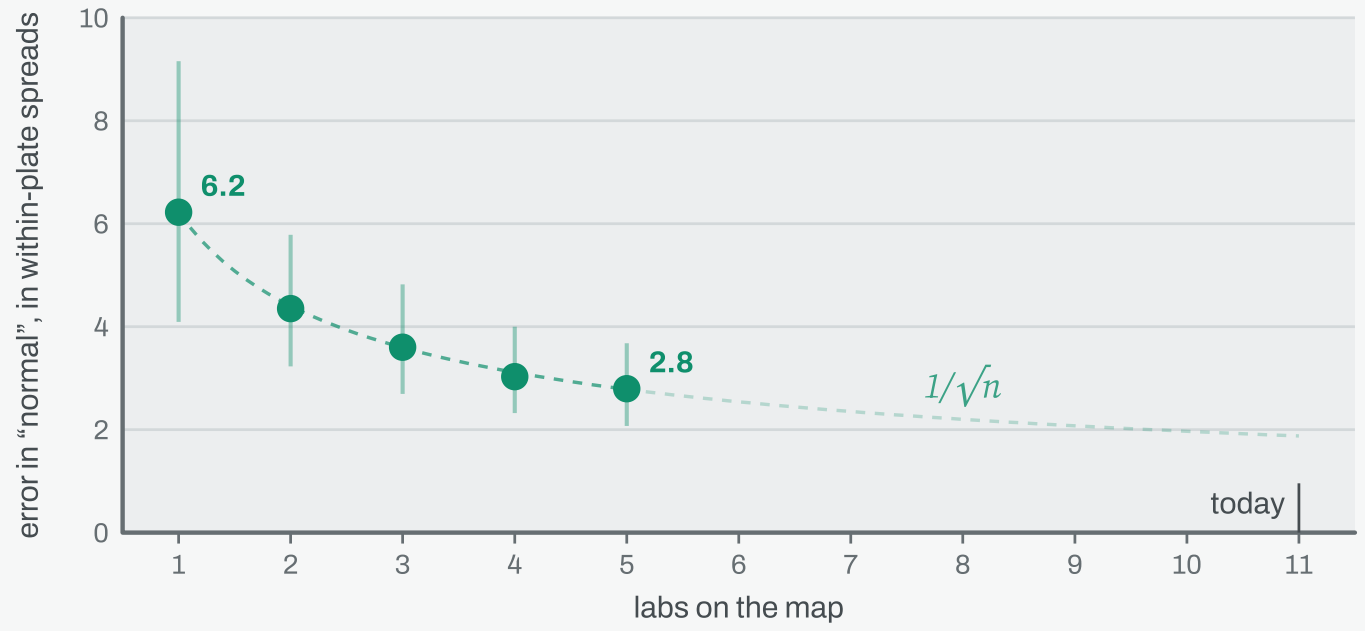
ONE PRICE

$30M

one cell map [11]

fig 5 EU-OPENSCREEN · one protocol · Hep G2 · as reported [13] · A23

Every lab that joins sharpens “normal” for everyone.



ERROR IN “NORMAL”

6.2 → 2.8

one lab → five labs on the map

MAP today

↓

CERTIFICATE next

↓

EXCHANGE then

↓

JOURNALS raw images, every paper

↓

…

fig 6 two maps, different labs · 1,871 plates · JUMP [7]

Drop in your images.
Ask anything.



Ask your data

Do my controls look “normal”?

yes · in distribution

Which lab is closest to mine?

lab 6 · 4.4 away

Do my drug results match other labs?

0.70 → **0.77** on the map

Did my cells grow as usual?

188 per well · in range

questions an agent suggests · proposed

fig 7 replayed on public data · JUMP [7] · A11

Reliable data ranks first.
Your rank is yours to share.

LEADERBOARD · U2OS

controls only

RANK	LAB	BATCH	PLATES	FIELD SCORE
#1	lab 5	28 Jun	7	82
#2	lab 5	2 Jul	6	79
#3	lab 5	6 Jul	5	79
#4	lab 5	16 Jul	8	78
#5	lab 5	1 Oct	6	77
…				
#48	**lab 2**	2 Aug	18	your best 58
…				
#95	**lab 2**	16 Aug	25	this batch 36

Your standing

#95 of 129 batches

YOUR STREAK

field best 33

TO CLIMB · DRIFT

you	2.9
field	1.2

fig 8 score = 100 − mean percentile · untreated wells · JUMP [7] · A17

Your data, certified.
Ready to share.

EXCHANGE · U2OS CONTROLS

DATASET	PLATES	BATCHES CERTIFIED	CONVERTS
lab 3	237	■■■■■□■■■□■■	0.80 → **0.81**
lab 6	225	■■■■■■■■■■■■	0.71 → **0.76**
lab 2	219	■■■■■■■■■□□□	0.75 → **0.80**
lab 10	205	■■■■■■■■■■■■■■■■	0.80 → **0.81**
lab 8	199	■■■■	0.84 → **0.88**
lab 5	197	■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■	0.81 → **0.84**
lab 11	172	■■□■■	0.70 → **0.73**

… 4 more

Your listing

10 of 13 batches certified

PROVENANCE

every well named lab 2 by its own cells

ACCESS

by subscription · lab keeps a share · proposed

fig 9 hollow: flagged · replayed · JUMP [7] · terms proposed · A23, A24

Six questions, six tools.

		DOES NOT SEE
STR authentication	Is it the line the label says?	behaviour
vendor certificate	Was it fine when it shipped?	after shipping
plate controls	Did this plate work?	whole lab drifting
phenomics platforms	Is my own data consistent?	lab differences, by design
data factories	Where do I get cell data at scale?	other labs' "normal"
Datum Line	**Where do my cells sit, against every other lab?**	which line · beyond fluorescence, yet

use all of them · A3, A7, A23

Eighteen months, four milestones.

$1.8M A21

1

benchmark playground

U2OS, A549, HepG2, iPSC · widefield and confocal · reference drugs · A16

2

three labs on the map

U2OS first · any line

3

model for any image

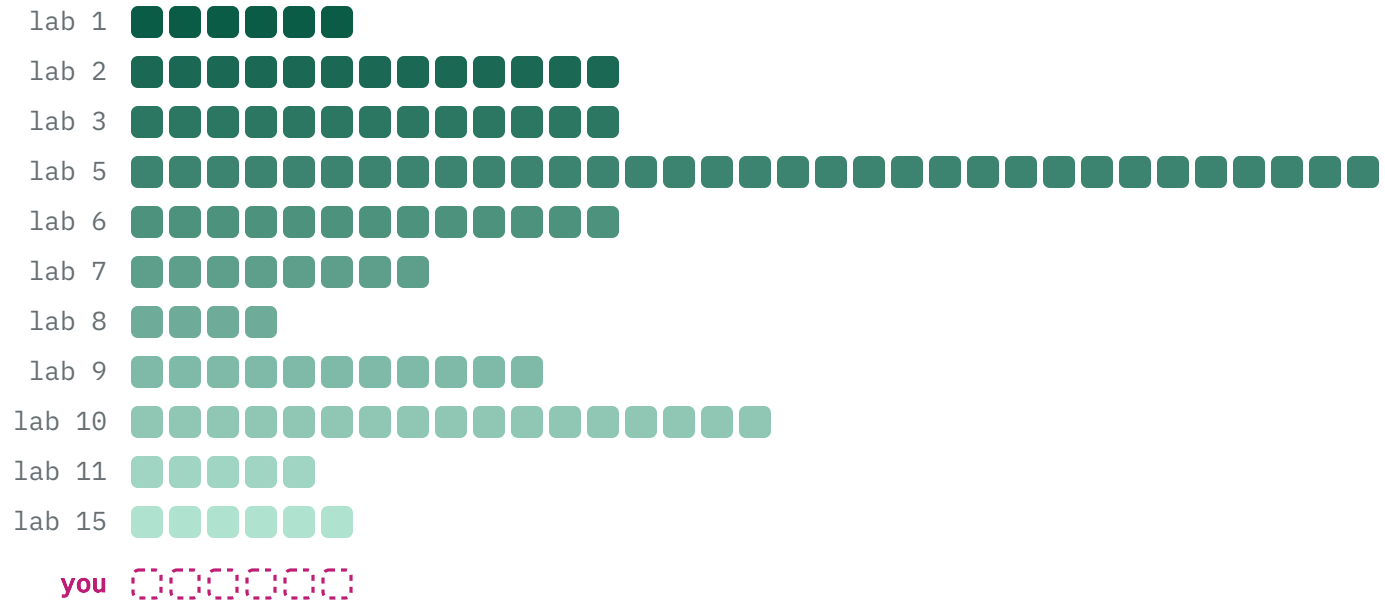
99.6% from one brightfield image · A19

4

first paid certificate

proposed

Every plate you image, placed on the map.



MAP → today

CERTIFICATE → EXCHANGE → JOURNALS → raw images, every paper → …

WHAT IT NEEDS

- leader
- bench
- funding
- partners

Have control images? Talk to me.

fig 10 one square per batch · 1,871 plates · JUMP [7] · A15, A24

The rest, if you ask.

Every number comes from the code here or a cited source.
Unfinished work is marked pending.

A1 **Why not trust the plate's controls?**
sick plate, sick control

A2 **What is the math?**
six calculations on 3,773 numbers

A3 **Why not batch-correct it away?**
erases most of the lab fingerprint

A4 **Is the model the moat?**
counting cells often wins

A5 **Why not just count cells?**
same count · different texture

A6 **How often do I re-certify?**
your baseline expires

A7 **Isn't this cell-line DNA testing (STR)?**
which cells · how they behave

A8 **Is the assay itself working?**
known answer

A9 **Does it change the result?**
same drug · different answer

A10 **Does it work without stains?**
bet on brightfield

A11 **Does it flag too often?**
six of 129 batches

A12 **What moved?**
stain brightness, not cell count

A13 **How does one lab move over time?**
lab's check · field's view

A14 **What does the map buy?**
same drug, closer between labs

A15 **Who pays?**
labs free · buyers of cell data pay

A16 **What would the money build?**
ugliest dataset · on purpose

A17 **Is the leaderboard fair?**
controls only, never results

A18 **What is private?**
control wells leave · nothing else does

A19 **What is the machine learning?**
image in · position and correction out

A20 **What about biosecurity?**
cells cannot lie about their origin

A21 **How much, and what for?**
planned $0.9M · asked $1.8M

A22 **How big?**
count × price · three streams

A23 **What is cell data worth?**
priced in deals · sites disagree

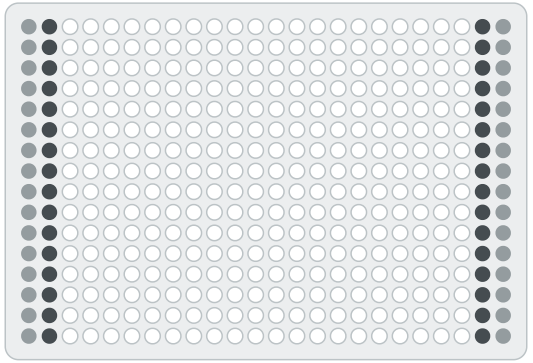
A24 **Why would a lab share?**
owned by the lab · verified by anyone

A25 **How do labs check cells today?**
search, ask, guess

Sick plate, sick control.

Controls sit on the same plate as the tests.

A REAL JUMP PLATE · 384 WELLS



untreated 32 · known effect 32 · test 320



test = *normal* + *plate* + *effect*

control = *normal* + *plate*

difference = *effect on this plate*

plate cancels from the number, never from the cells

Seeing a sick plate takes a view from outside it.

Fig. A1 · left: JUMP plate 1053601756, lab 2 · right: schematic, no measurements

Six calculations, one well.

Each well becomes 3,773 numbers.
Each result compares averages, or tests a model on unseen plates.

1 · IMAGE TO NUMBERS [1]

$$e = \varphi(image)$$

Three ways to turn images into numbers gave similar results on liver cells.
Tool is not the asset.

2 · WHICH LAB? 99.9%

$$f : e \to lab$$

Names the lab from one well, on plates it never saw.
Far above chance (9%): each lab leaves a fingerprint.

3 · HOW FAR FROM NORMAL? [e6]

$$d = \| E[e \mid plate] - E[e \mid all\ labs,\ c] \|_{\sigma}$$

How far this plate's average sits from all labs' average, in units of normal spread.
Direction says what moved.

4 · SOURCES OF VARIATION pending

$$Var(e) = V_{model} + V_{lab} + V_{batch} + V_{plate} + V_{well}$$

Splits variation into microscope type, lab, batch, plate, well.
Shares add up to 100%.

5 · REMOVE KNOWN CAUSES 24%

$$e_{\perp} = e - \hat{E}[e \mid c]$$

Remove only causes known to be the instrument.
Subtracting each plate's controls goes further: right lab only 24% of the time.

6 · DID THE DRUG CHANGE THE CELLS?

DISTANCE $\delta = b(\| e - E[e \mid untreated] \|)$

CLASSIFIER $g : e' \to label$

Distance from the untreated average, capped so one odd well cannot dominate.
Or a model, after aligning to the reference.

drug discovery · remove the lab first
quality check · keep the lab: it is the signal

What JUMP records. What it leaves out.

What counts as normal depends on every variable below.

RECORDED IN JUMP

lab · batch date · plate · plate type · well position · microscope · confocal or widefield · laser or LED · objective NA · fields per well · filter set · pixel size · brightfield planes · CellProfiler version

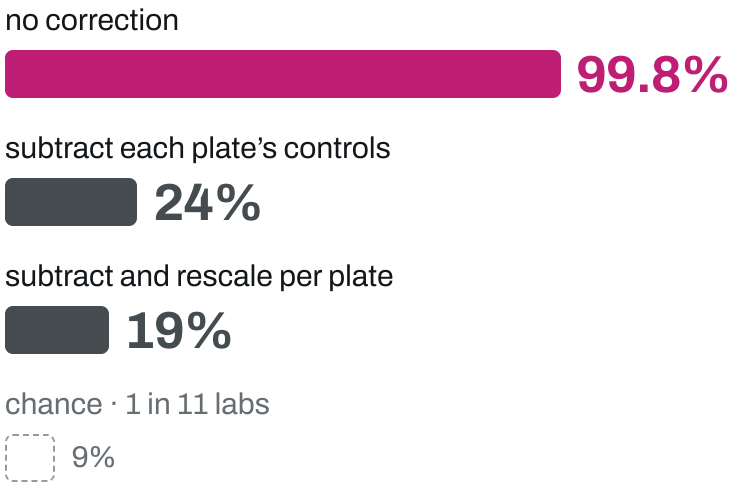
in every image file too:
image size · stored bit depth

NOT IN ITS PUBLIC METADATA

DMSO % per well · DMSO lot · DMSO water uptake · stock freeze–thaw · serum lot · media lot · passage · thaw date · seeding density · confluence at dosing · dosing to fixation · stain and dye lots · operator · incubator shelf · edge evaporation · mycoplasma status · CO2 and humidity

not required to publish today [15]
in our own data: tracked, every one

CAN A MODEL TELL WHICH LAB?
HIGHER = CLEARER LAB FINGERPRINT



29,943 untreated wells, 16 per plate
tested on plates it never saw

Standard first step: subtract each plate's own controls.
It erases most of the lab fingerprint a certificate reads.

Batch correction is built to remove this signal: Arevalo et al., Nat Commun 2024 · variables: JUMP metadata [7]

NATURE COMMUNICATIONS · 2026 · [2]

Weak benchmarks: counting cells often wins.

I coauthored this paper.
Benchmarks decide which model looks good.

… counting cells achieves unexpectedly high performance in these benchmarks, making them less useful for discerning whether additional properties, such as phenotypic profiles (mRNA or Cell Painting), provide additional useful information on bioactivity. … we recommend filtering benchmarks to exclude such assays and including a cell-count baseline. … we confirm that models leveraging Cell Painting image-based profiles outperformed the baseline cell count model. …

ABSTRACT EXCERPT · VERBATIM `finding` `recommendation · what held up`

209 and 270

tests in two popular benchmarks
many really measure cell health

Images win

on 24 tests aimed at one protein each
Cell Painting beat the cell count

WHY IT MATTERS HERE

Whoever sets the baseline sets the benchmark.

Datum Line benchmark · cell-count baseline on every task · planted failures · every model graded · A16

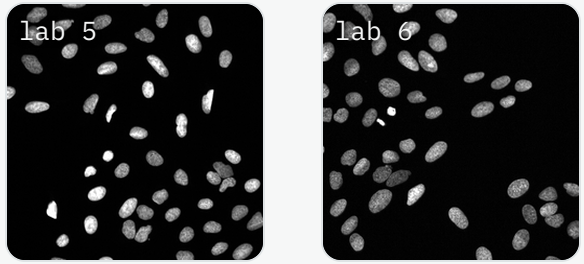
Seal S, Dee W, … **Titterton K**, … Carpenter AE. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nat Commun*, 2026.

CELL SYSTEMS · 2026 · [1]

Same count.
Different texture.

Count moves only when cells die or divide.
Image moves when anything changes.

SAME CELL LINE, SAME PROTOCOL



196	≈	222	count
0.94	≠	1.10	texture

count gap 0.4× usual spread · same
texture gap 2.4× usual spread · different
usual spread: plate to plate, within one lab
texture: % of DNA signal in fine detail

METHOD 1 · CLASSIFIER

Does this run belong with the past runs?

standard alignment · our tuning

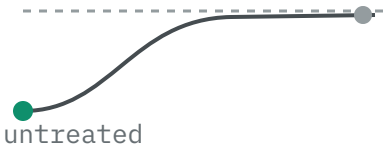


removes day-to-day shifts

METHOD 2 · DISTANCE

How far out, and which way back?

distance published [1] · transform unpublished



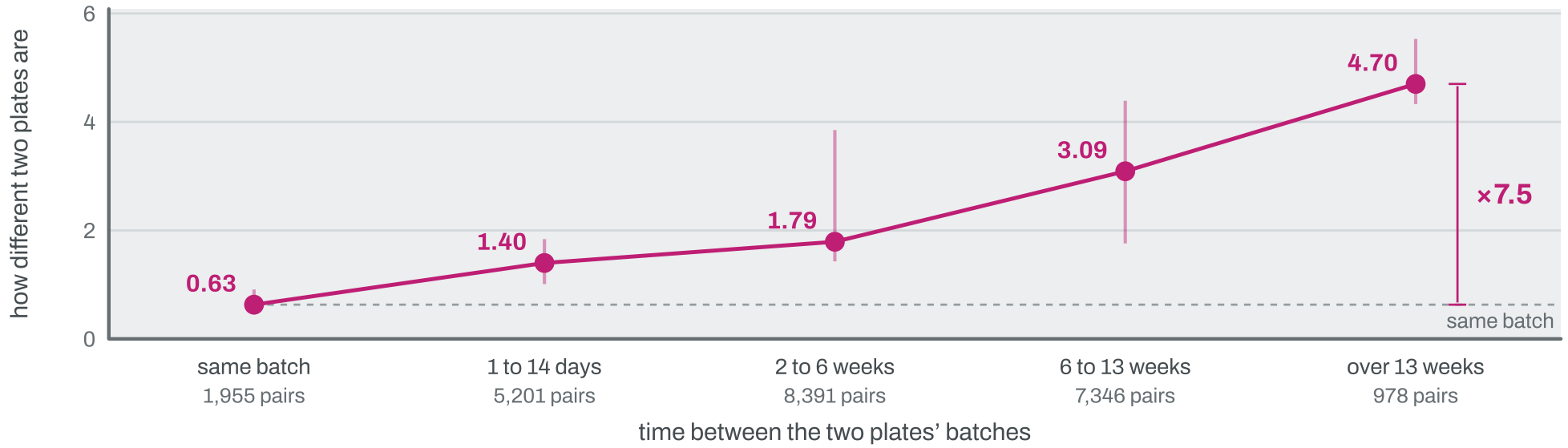
one odd well cannot dominate

both need a reference · somebody else's images

Fig. A5 DNA channel · JUMP cpg0016 [7], CC0 · nuclei by Otsu · texture = high-frequency energy over total, inside nuclei · results/a5-pair

Your baseline expires.
Re-check at least every quarter.

Plates of one lab drift apart with time.
Still rising after 13 weeks.



IS THE CELL LINE RUINED?

This data cannot tell.
It shows when the lab moved, not why.

HOW OFTEN?

At least every quarter.
By 6 to 13 weeks: ×4.9 vs same batch.

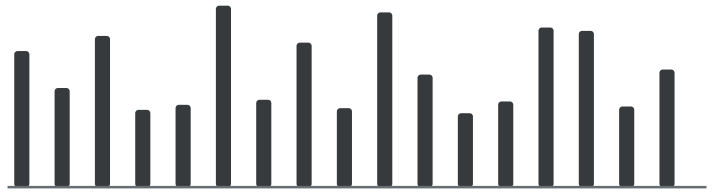
HOW SOLID IS THE NUMBER?

One lab, 219 plates, 13 batches.
Other labs not yet measured.

Dot: median plate pair. Bar: middle half of pairs.
1 = how far a typical untreated well sits from its plate's average.

STR says which cells.
Images say how they behave.

STR · DNA FINGERPRINT

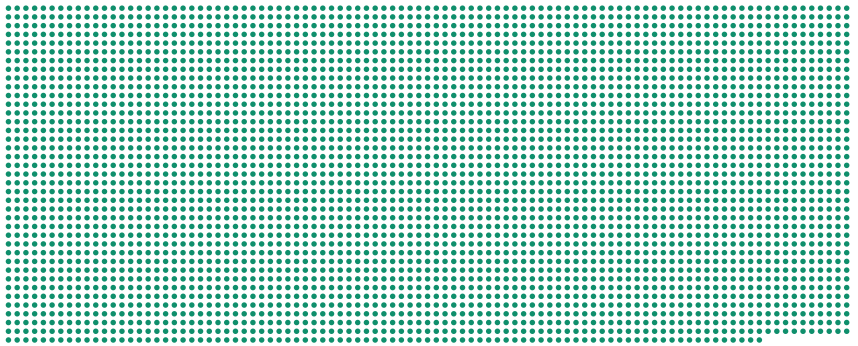


17
DNA markers

once
per cell line

Standard test for identity.
Cannot say how the cells behave.

DATUM LINE



3,773
numbers per well

every batch
images already taken

Shows how cells behave, next to other labs.
Cannot say which cell line it is.

identity · STR + behaviour · Datum Line = **one certificate**

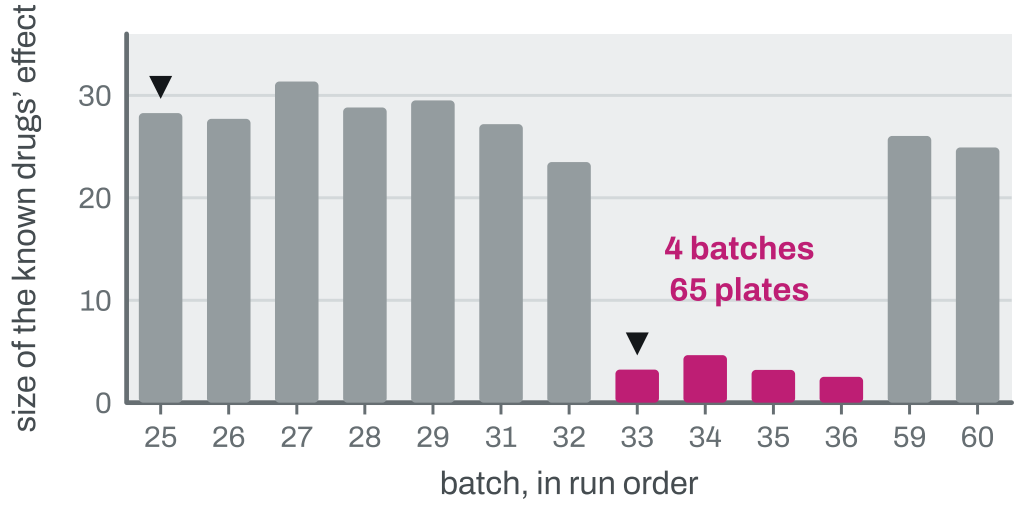
natural first partner: STR testing services

STR: ATCC [14], 17 loci plus amelogenin. Datum Line: 93,228 untreated wells, 1,872 plates, 11 labs, JUMP [7].

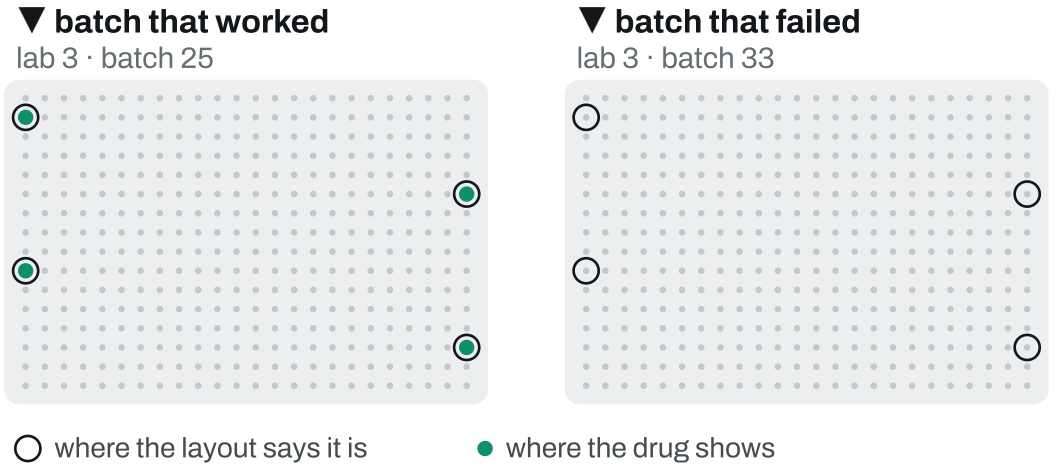
Known answer, run in every lab.

Untreated wells show where a lab sits.
Known drugs show whether the test works.

LAB 3 · DID THE KNOWN DRUGS WORK?



WHERE DOES THE KNOWN DRUG SHOW?



125 of 129

PUBLIC BATCHES · KNOWN DRUGS WORKED

1 of 4

FAILED BATCHES CAUGHT BY UNTREATED WELLS · USE BOTH CHECKS

Failed: under a quarter of the lab's usual effect, and unlike other labs. 1,829 plates, JUMP [7]. Plates C13443aW, A12069aW: JUMP plate layout.

Same drug.
Different answer.

8 known drugs, 11 labs.
Each drug's effect changes from lab to lab.

WHAT A DRUG SCREEN REPORTS drug wells minus untreated wells, on the same plate

HOW WELL A PLATE AGREES ON WHAT A DRUG DOES

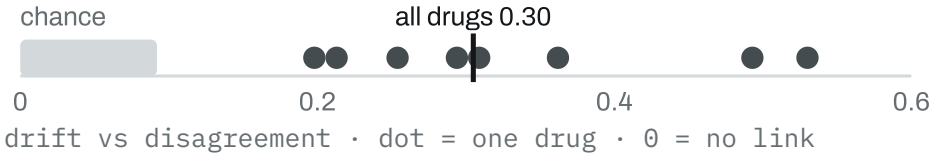


Q1 · SAME ANSWER IN ANOTHER LAB?

Less alike than in the same lab, for all 8 drugs.

Q2 · DO UNTREATED WELLS WARN YOU?

Inside a lab, partly.
Plates that drift disagree more.



Q3 · DOES IT MATTER?

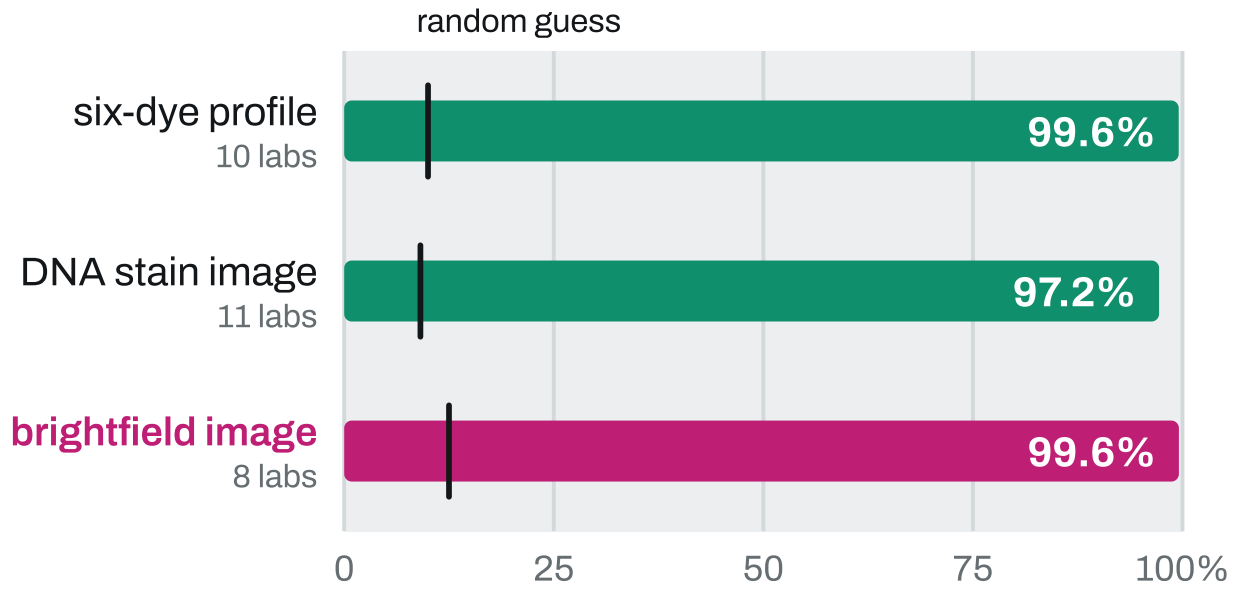
Which lab ran it explains 20% to 71%
of the spread in effect strength.

8 known drugs · 1,829 plates · 11 labs · JUMP [7]

Shown in fluorescence.
Bet on brightfield.

Evidence: cells stained with six dyes.
Product: plain light, no dye.

CAN ONE WELL TELL WHICH LAB IT CAME FROM?



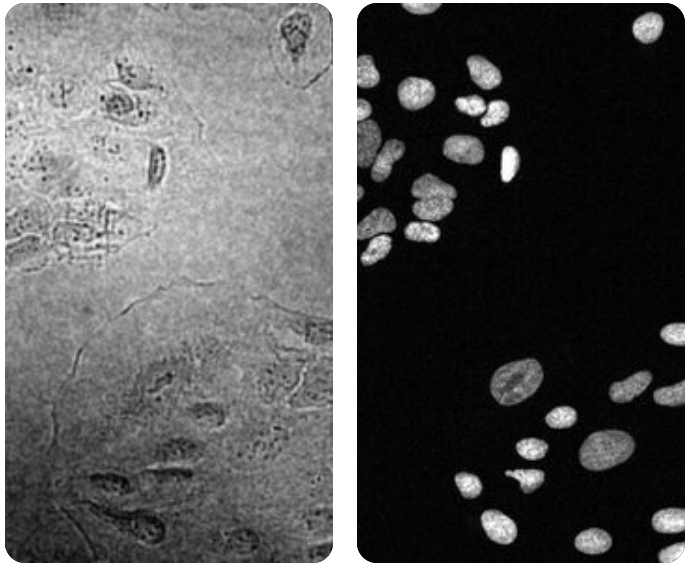
Brightfield already tells labs apart.

Not yet the whole map: nearest lab matches the dye map in 29 of 102 batches.

WHY THE BET IS FAIR

No dye: fewer things can go wrong

~~stain lot~~ · ~~dye lot~~ · ~~staining step~~ A3



BRIGHTFIELD DNA STAIN

same spot in one well · lab 2

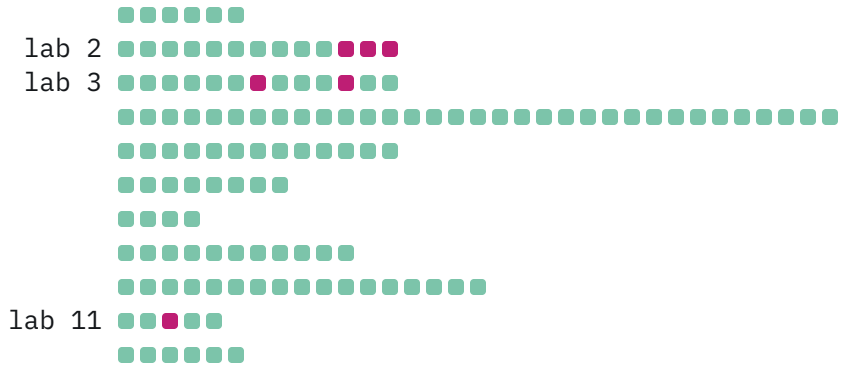
% named right, each lab counted equally · tested on unseen batches · one untreated well · image model never trained on cells · JUMP [7]

Does it flag too often?

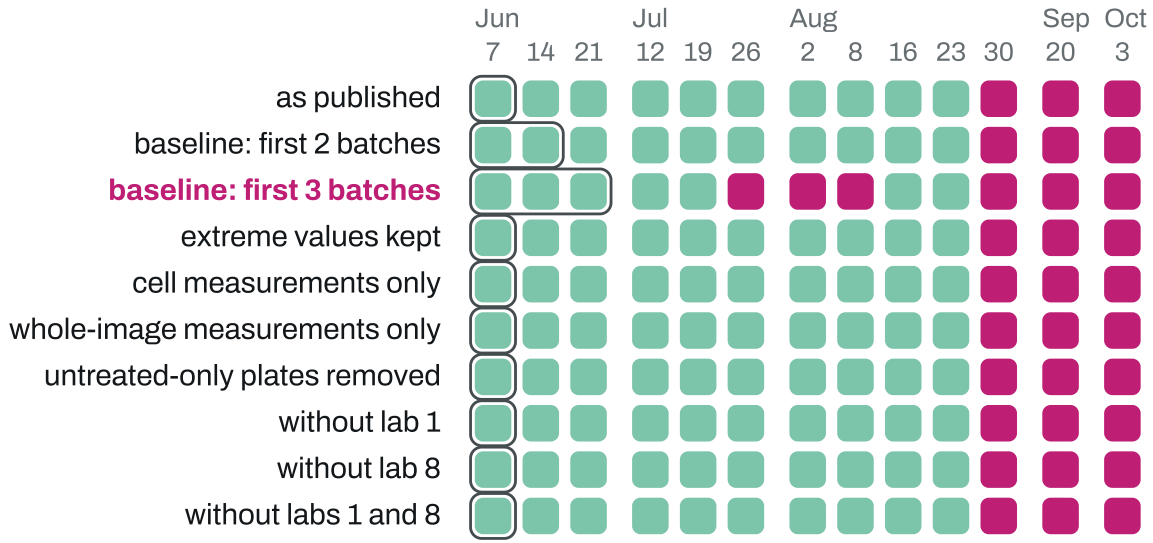
6 of 129 batches.

Rare: 8 of 11 labs never flagged.
Lab 2: same result 9 of 10 ways of checking.

ALL 11 LABS · 129 BATCHES



LAB 2 · 13 BATCHES, CHECKED 10 WAYS



WHAT A FLAG MEANS

Batch looks more like another lab
than like its own first batch

LAB 2 · PLATES FLAGGED

25 of 25 in flagged batches

2 of 194 in the other batches

WHAT CHANGES IT

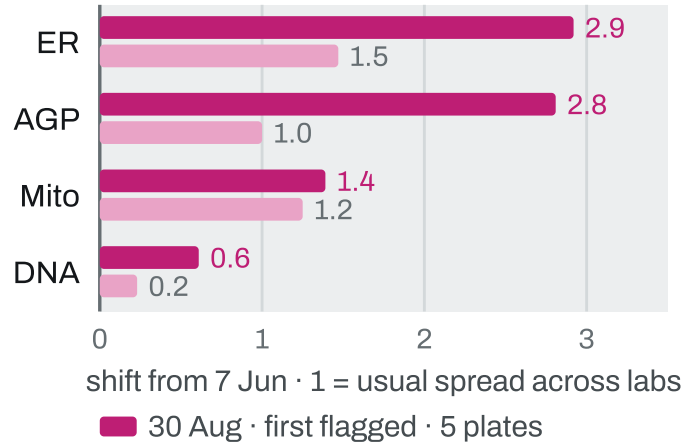
1 of 10 ways differs
its baseline includes 21 Jun, a batch with fewer cells

■ not flagged ■ flagged ▢ baseline

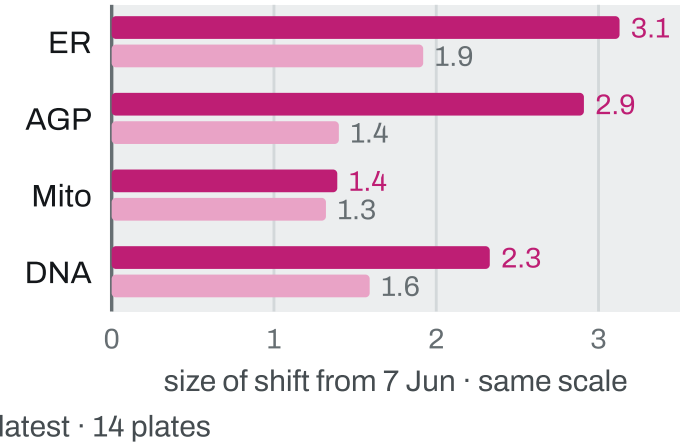
Brightness rose.
Cell count did not.

Lab 2's flagged batches against its first batch, 7 Jun.

BRIGHTNESS



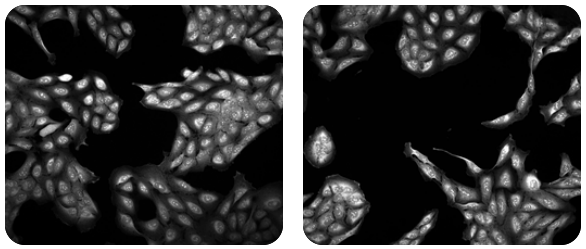
PATTERN (TEXTURE)



30 Aug · first flagged · 5 plates

3 Oct · latest · 14 plates

ON SCREEN, BOTH LOOK FINE



7 JUN · ER
7,612

30 AUG · ER
51,230

raw brightness 7× higher · DNA stain 20×
ER pattern: around the nucleus → whole cell

IN NUMBERS

ER and AGP brightness:
96% or more of measurements rose
cells per well: 189 (7 Jun) → 165 (30 Aug) → 182 (3 Oct)
inside the range of its earlier batches, 150–199

CAMERA OR CELLS?

Cell measurements moved, not only whole-image ones.
This check cannot tell a new stain batch from real biology.

WHAT A LAB GETS

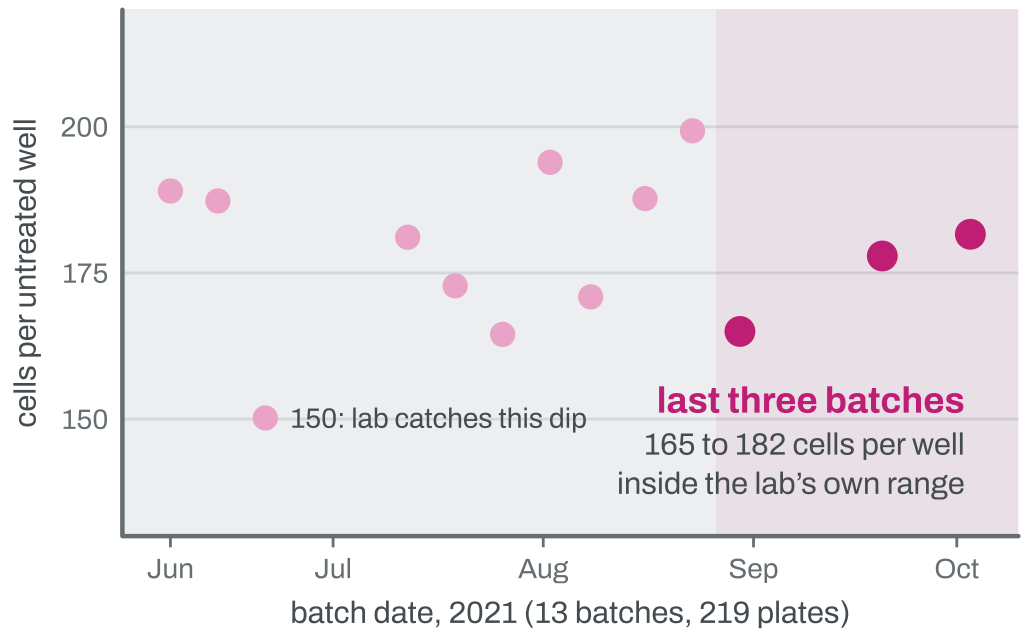
Look first: camera exposure, ER stain
Then: stain mix-up,
colour leak between stains, or biology
Finding the cause is the lab's job

ER, AGP, Mito, DNA: four stain channels, each marking different parts of the cell · each bar averages one channel's measurements · JUMP [7]

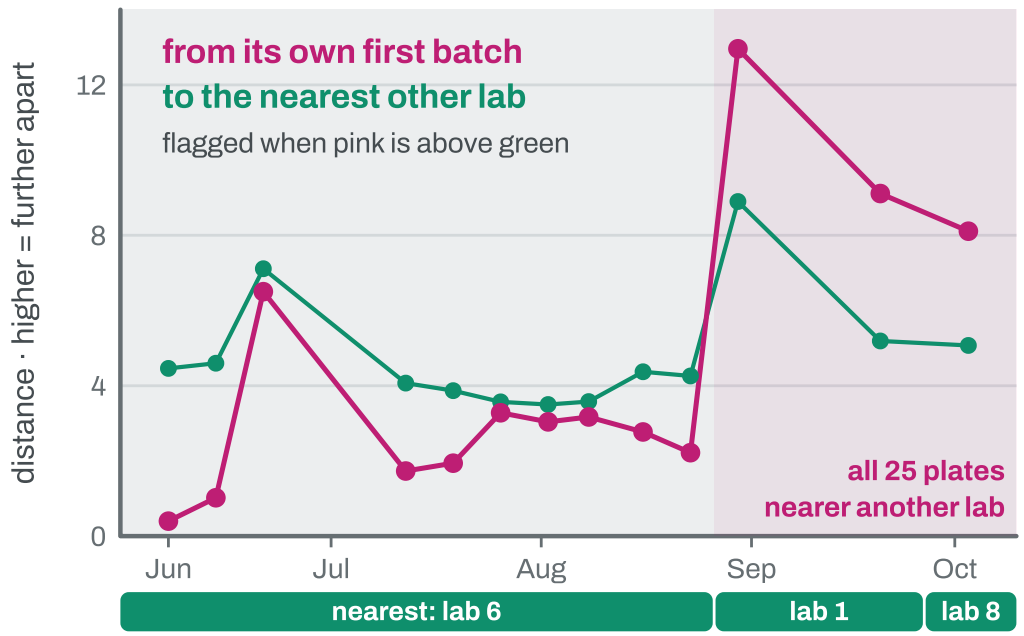
Lab 2's own check: fine.
Next to other labs: it moved.

Same 13 batches, two views.
Not a verdict: a reason to look.

LAB'S OWN CHECK: CELLS PER WELL



COMPARED WITH OTHER LABS



distance unit: usual spread between wells on one plate · lab 2 · 2021 · JUMP [7] · what moved: A12

Same drug, two labs. Closer on the map.

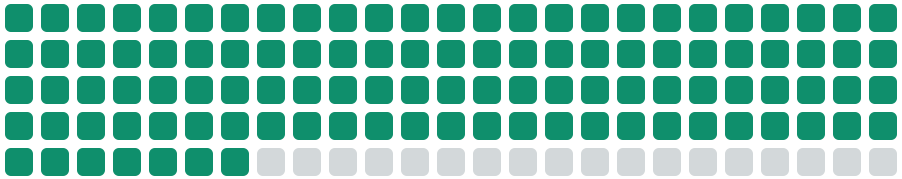
Seven drugs with a known effect, 125 batches, 11 labs.
Higher is better.

Do labs agree on what a drug does?

0.75 → **0.76** → **0.78**

raw · usual fix · on the map

1 = every lab sees the same effect



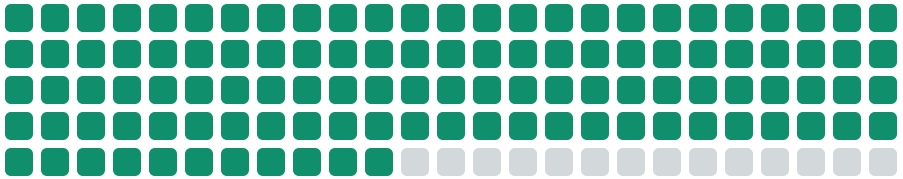
107 of 125 batches closer on the map than with the usual fix

Can you still tell the drugs apart?

0.52 → **0.56** → **0.59**

raw · usual fix · on the map

gap to the nearest wrong drug · higher is easier



111 of 125 batches closer on the map than with the usual fix

NOTHING LOST

Right drug recognised: 0.92 → 0.94
Matches its own past: 0.88 → 0.88

HOLDS

Labs agree more: 5 of 5 settings
Drugs stay apart: 4 of 5

NOT FIXED YET

A drug's strength varies batch to batch
20% usual fix · 12% map · 6% raw

Usual fix: each batch scaled to its own untreated wells. The map: scaled to every other lab's. 1,829 plates · JUMP [7]

PROPOSED · PRICES ARE THE ASSUMPTIONS ON A22

Labs contribute free.
Buyers of cell data pay.

Certificates on bought data pay first.
Then a data exchange grows from them.

WHO	WHAT THEY GET	WHAT THEY PAY	WHAT IT BUILDS
labs and core facilities	a place on the map, the certificate, the leaderboard	**free, for contributing controls**	the map
companies that buy cell data	a certificate on every dataset they buy	**~$10–50k per certificate**	first revenue
teams that run cell screens all year	a check on every batch they run	**~$5–15k per cell line, per quarter**	repeat revenue
AI labs and model builders	a test set with faults planted on purpose	**~$50–200k a year, as partners**	the standard
then: the exchange	access to certified data from many labs	**a subscription; each lab gets a share**	the network

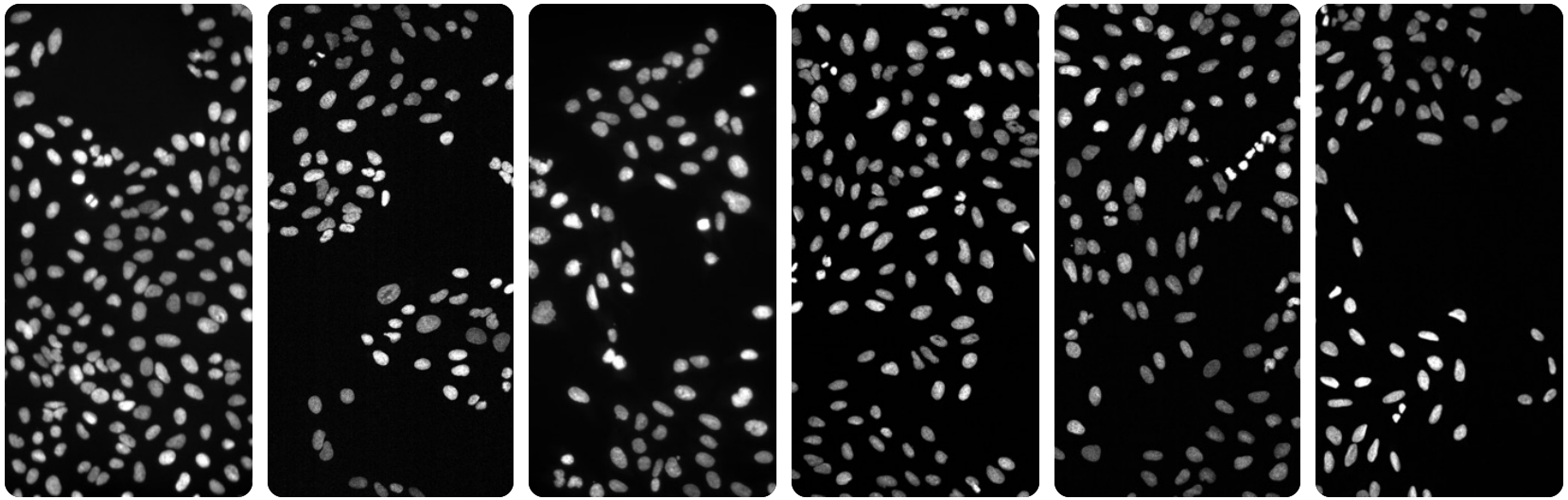
Labs send control wells only. Nothing else leaves the lab.

Build the ugliest dataset in biology, on purpose.

Same cells, same protocol, no drug.
Still, each lab looks different.

PUBLIC DATA TODAY · SIX LABS

real wells · JUMP [7]



LAB 1	LAB 2	LAB 4	LAB 5	LAB 6	LAB 10
Phenix · widefield	CV8000 · confocal	Phenix · widefield	CV8000 · confocal	CV8000 · confocal	CV8000 · confocal

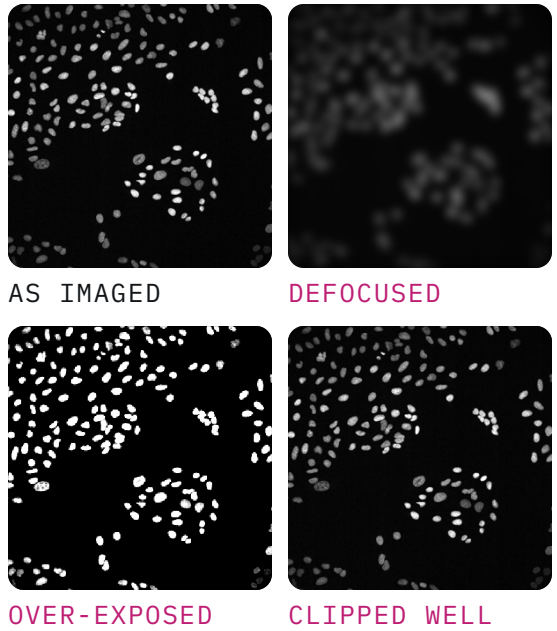
5 microscope models

8 plate layouts

13 labs [7]

PLANTED ON PURPOSE

simulated



AS IMAGED

DEFOCUSED

OVER-EXPOSED

CLIPPED WELL

Can a model name the fault?

playground · U2OS, A549, HepG2, iPSC [17] · widefield and confocal · known drugs · planted faults · every correction scored

A game, not a report card.

Every batch ranked on its untreated wells, never on results.

LAB	BATCHES	SCORE	BEST RANK	CLEAN RUN	HOW UNUSUAL	PLATES DIFFER	BATCH TO BATCH	CELL COUNT OFF
					lower is better · green: better than the all-labs row			
lab 5	33	**70**	#1	33	3.35	0.27	0.81	0.31
lab 6	13	**60**	#16	13	4.44	0.41	1.23	0.12
lab 10	17	**54**	#10	17	3.84	0.62	1.21	0.23
lab 8	4	**52**	#54	4	4.10	1.08	0.86	0.09
lab 11	5	**47**	#23	2	3.79	0.87	1.57	0.17
lab 3	13	**44**	#18	6	3.27	0.80	2.01	0.29
lab 7	8	**42**	#69	8	6.38	0.58	1.17	0.33
lab 2	13	**39**	#48	10	5.68	0.57	1.99	0.14
lab 1	6	**32**	#78	6	8.68	1.10	2.03	0.09
lab 9	11	**25**	#97	11	9.97	0.50	1.76	0.45
lab 15	6	**23**	#68	6	5.53	0.87	1.69	0.44
all labs	**129**	—	—	—	**4.01**	**0.51**	**1.17**	**0.28**

SCORE

0 to 100 · higher is better

from the average rank on the four columns
among 129 batches

BEST BATCH

82

lab 5 · 28 Jun · 7 plates
lab 5 holds 9 of top 10

YOUR CHOICE TO PLAY

Named only if you opt in.
Private version for your own plates.

Lab medians · 1,871 plates · 11 labs · clean run: batches in a row still closer to the lab’s first batch than to any other lab · JUMP [7]

Control wells leave.
Nothing else does.

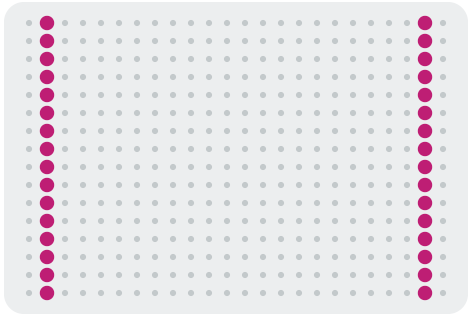
No compound, no target, no result.
Nothing for legal to approve.

WHAT LEAVES THE BUILDING

untreated control wells only

images, or the numbers measured from them.
No compound.

a plate map of the control wells



pink: 32 of 384 wells leave · JUMP [7]

WHAT DATUM LINE DOES WITH IT

builds the map

one average point per plate.
Scale set by the other labs, never by yours.

the map holds untreated wells only

82,319

for lab 2: untreated wells of 10 other labs
0 treated

answers

where you sit · how far you moved · nearest lab · cell count · normal or not · rank

sends a correction, not a model of you

one per batch, fitted on your untreated wells · A19

WHAT IT NEVER SEES

your treated data

the correction is sent to you.
Your results are corrected in your building, by you.

your name, unless you choose

proposed: labs appear as a number; a lab opts in to a name. Every JUMP lab here is a number.

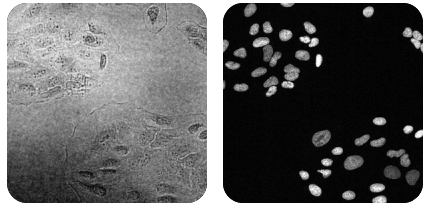
your images, once measured

proposed: measured on arrival, then deleted.
Only the numbers are kept.

Image in.
Position and correction out.

Results stay in the lab.
Untreated wells and the correction travel.

STEP 1 — AN IMAGE



no stain | DNA stain

One untreated well.

STEP 2 — NUMBERS

3,773

numbers per well today

Standard software today.
AI image model next:

no stain · 8 labs	99.6%
DNA stain · 11 labs	97.2%

right lab, from one picture

STEP 3 — THE MAP

99.9%

a computer names the lab of each well · by guessing: 9%

Every lab looks different.

Each is placed on the other labs' scale, never its own.

STEP 4 — THE CORRECTION

your batch → the map

one per batch · fitted on your untreated wells

Sent to you.

You apply it to your results.

Correction, per batch on 200 components: $x' = C_{field}^{1/2} C_{batch}^{-1/2} (x - m_{batch})$ · every script: github.com/kat-titter/datum-line/analysis

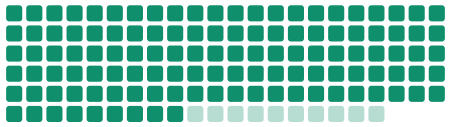
Cells cannot lie about where they came from.

One map of untreated wells, no results stored.
Two checks measured, two proposed.

PROVENANCE

99.9%

of 93,228 untreated wells get the right lab, on plates the computer had never seen.



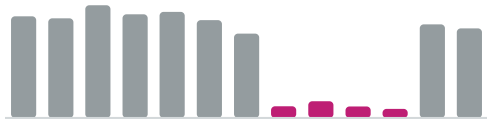
1 square = 1 batch
solid: every well right · 119

INTEGRITY

4 batches

Label: known drug in these wells.

Cells: almost no effect, on 65 plates.



lab 3 · drug effect per batch
pink: failed · A8

CELLS FROM NOWHERE

9×

The farthest of 129 real batches from its nearest other lab.

Farther out: maybe the wrong cell line, or a mixed culture.

untested · proposed as the first partner-lab test



distance to nearest other lab
1 line = 1 real batch

EARLY WARNING

proposed

Untreated wells are an early warning.

If they move and nobody changed anything, something in the room did: a reagent batch, the air, the water.

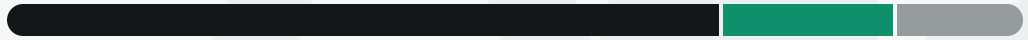


lab 2 · change since batch 1
filled: flagged · A13

Integrity check uses each plate’s known-drug wells (A8) · × = times the usual spread of wells within one plate · JUMP [7]

About $1.8M for eighteen months.

$884k planned + $884k buffer = $1.77M asked



people 71% · benchmark 17% · other 12%

leader, to run it	18 months	**$180k**
scientific lead (me)	18 months	**$180k**
ML engineer	from month 7	**$160k**
payroll overhead	20% of salaries	**$104k**
benchmark playground	~60 plates · planted failures · partner labs or a contract lab	**$120k**
partner-lab bench time	3 labs · controls they already image · reagent top-ups · shipping	**$30k**
compute and storage	images · models · hosting · API	**$45k**
legal, IP, insurance, accounting	incorporation to first contract	**$45k**
travel and partner visits	2 benchmark partners · 3 labs	**$20k**
planned	sum	**$884k**
+ buffer	most things take 2–3× the time and money planned · we plan for 2×	$884k
asked · 18 months		**$1.77M**

NO IN-HOUSE LAB

Partner labs or a contract lab.
Own lab: costs more, proves less.

WHAT THE MONEY MUST SHOW

1 benchmark playground, paid in this round · **2** three labs on the map
3 model for any image · **4** first paid certificate → seed

salaries below market · leader and scientific lead paid · nothing here committed

About $84M a year, if everyone joined.

Count × price.
Three streams.

assumption	low	base	high
organisations testing drugs on a few cell lines	200	400	800
cell lines each one uses per quarter	2	4	8
price per cell line per quarter, $k	5	10	15
organisations testing drugs on hundreds to thousands of cell lines	10	25	50
site licence per year, $k	50	150	300
subscriptions and site licences, per year	**$8M**	**$68M**	**$399M**
companies buying cell data they did not make	50	150	300
purchased datasets each certifies per year	2	4	8
price per certificate, $k	10	25	50
certificates on purchased data, per year	**$1M**	**$15M**	**$120M**
benchmark partners	4	10	20
membership per partner per year, $k	50	100	200
benchmark membership, per year	**$200k**	**$1M**	**$4M**
total per year, if every buyer joined	**$10M**	**$84M**	**$523M**

BASE CASE · PER YEAR



↑ five-year target · a tenth · $8M

benchmark $1M ↑

WHY BUYERS PAY

One drug screen I ran on cells that had changed: about $2.6M to redo.
Certificate: a small part of that.
Subscription: insurance against it.

prices proposed on A15 · academic labs join free · they are the map, not the revenue

Data has a price.
Cell data needs a standard.

Three public prices.
Same cells, four sites, different results.

AI TRAINING DATA

$1.5M a day

paid to experts for AI training data, by one vendor

~$85 an hour · $10B valuation [10]

CELL MAPS

$30M

paid by Roche for one cell dataset

second milestone · $213M to date [11]

CONSORTIA

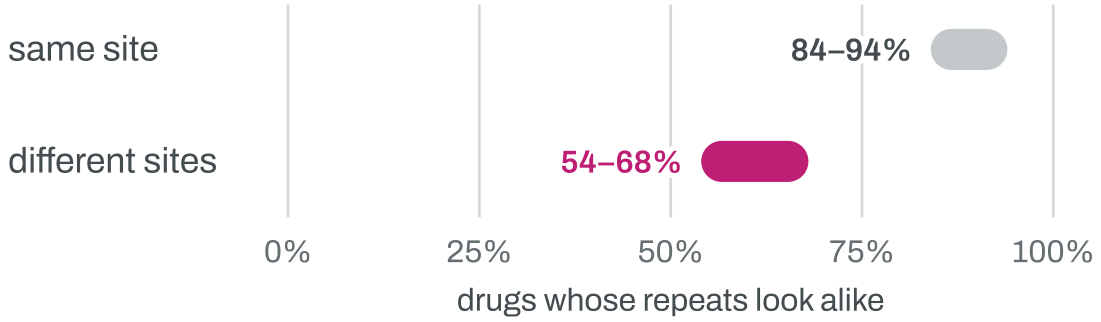
$1.8B

pledged for cell data to train AI
funders get it a year early

Meta, DeepMind, Isomorphic: $300M [12]

4 EUROPEAN SITES · ONE PROTOCOL · ONE CELL LINE

same cells, same reagents, shipped to every site [13]



WHAT THE PAPER SAYS

“not directly comparable between imaging sites” [13]

on one shared map, labs agree more · A14

[10] TechCrunch, 2025 · [11] Recursion, 2025 · [12] eWeek, 2026 · [13] Wolff et al., *iScience*, 2025 · full references in the README

Owned by the lab.
Verified by anyone.

Lab keeps its data and a share of each subscription.
No crypto token.

UNTREATED CELLS ONLY — today

no drug · no target · no result
results never leave the lab

LABS LEAVE FINGERPRINTS — today

99.9%

right lab out of 11, from images alone

so anyone can check the source · A20

SIGNED AND TIMESTAMPED — proposed

public record · anyone can check
dates cannot be changed · blockchain optional

LAB KEEPS ITS DATA — proposed

listing is opt-in
withdraw any time

KNOWN BUYERS — proposed

verified at sign-up · odd access flagged
access log open to members

LOW RISK BY DESIGN — today

cell-line images only
no human genetic data · no pathogen data

1 **lab lists**
images of untreated cells
certificate attached

→

2 **certificate checks**
how normal the cells look
which lab · what changed

→

3 **subscriber pays**
for access, not ownership

→

4 **lab keeps a share**
of every subscription
as long as its data is used

nothing signed yet · privacy A18 · biosecurity A20 · prices A15

Today: search, ask, guess.

WHY NOW

- 2023 — public cell image set [7]
- 2025 — FDA: fewer animal tests [9]
- 2026 — the map

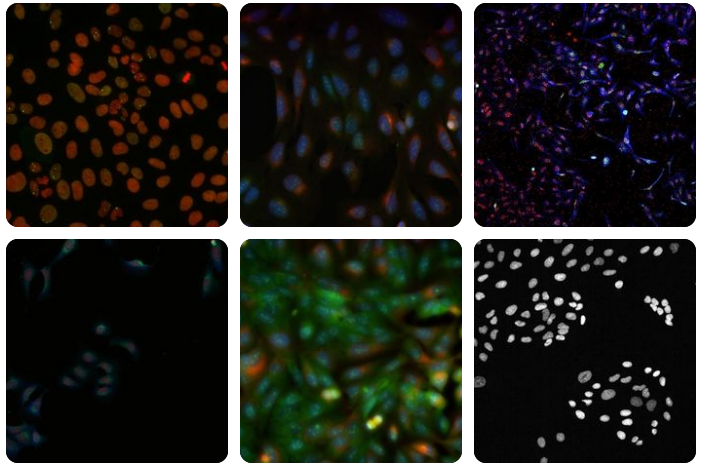
$28B A YEAR · SPENT ON US LAB RESEARCH OTHERS CANNOT REPRODUCE [6]

36% ≈ $10B · cells and reagents

64% · other causes

1
search

do my U2OS cells look normal

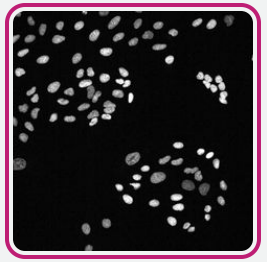


one cell line · six papers · all look different

2
ask

To: whoever ran this line last

Subject: does this look right to you?



screenshot.png

3
guess

$ RUN IT ANYWAY

hundreds of thousands

of R&D dollars at risk

WAIT TO BE SURE

months

of development lost

[6] Freedman 2015